5 MARKET OVERVIEW

5.1 INTRODUCTION

The microbiome sequencing services market is estimated to grow at a CAGR of 14.3% during the forecast period. Growth in this market is majorly driven by the increasing demand for metagenomic sequencing, the growing focus on human microbiome therapeutics development, and the high cost of advanced infrastructure for sequencing. However, a dearth of skilled personnel is expected to hinder the growth of this market during the forecast period.

5.2 MARKET DYNAMICS

FIGURE 18 MICROBIOME SEQUENCING SERVICES MARKET: DRIVERS, RESTRAINTS, OPPORTUNITIES, AND CHALLENGES

DRIVERS	▪ Increasing demand for metagenomic sequencing ▪ Growing focus on human microbiome therapeutics development ▪ High cost of advanced infrastructure for sequencing
RESTRAINTS	▪ Dearth of skilled personnel
OPPORTUNITIES	▪ Expanding applications of microbiome sequencing in therapeutics development
CHALLENGES	▪ Issues related to sample preparation/isolation of microbes for sequencing

Source: Secondary Literature, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 4 MICROBIOME SEQUENCING SERVICES MARKET: IMPACT ANALYSIS OF DRIVERS, RESTRAINTS, OPPORTUNITIES, AND CHALLENGES

FACTOR	IMPACT		
	2–3 YEARS	3–5 YEARS	5–10 YEARS
DRIVERS			
Increasing demand for metagenomic sequencing	High	High	High
Growing focus on human microbiome therapeutics development	Medium	High	Medium
High cost of advanced infrastructure for sequencing	Medium	Medium	Medium
RESTRAINTS			
Dearth of skilled personnel	High	Medium	Medium
OPPORTUNITIES			
Expanding applications of microbiome sequencing in therapeutics development	Medium	Medium	High
CHALLENGES			
Issues related to sample preparation/isolation of microbes for sequencing	Medium	Low	Low

5.2.1 DRIVERS

5.2.1.1 Increasing demand for metagenomic sequencing

Metagenomic sequencing is a target-independent approach that offers a comprehensive view of pathogenic agents and enables the detection of common and unexpected pathogens in samples. In recent years, there has been a significant increase in the demand for metagenomic sequencing in the field of microbiome research owing it its wide array of applications, such as disease diagnosis, drug discovery, and precision medicine. Recent developments in metagenomic sequencing technologies, such as improvements in sequencing accuracy, scalability, and data analysis, support the use of metagenomic sequencing. The development of long-read sequencing, single-molecule real-time sequencing, and nanopore sequencing has been particularly helpful in the metagenomic sequencing market. The demand for long-read sequencing-based metagenomics has increased in samples that are less complex, involve host-associated microbiomes, and comprise fewer microbial strains than microbiomes in marine environments, sediment, and soil. Such technologies are expected to drive the demand for microbiome sequencing services.

Over the years, the demand for precision medicine has increased. Precision medicine involves tailoring medical treatments and interventions to an individual's unique characteristics, including their microbiome. Metagenomic sequencing services are helpful in gathering essential information for personalized medicine, as well as identifying specific microbial signatures associated with disease risk and specific treatment responses. Metagenomic sequencing is also being used to study the role of microbiomes in cancer development, progression, and response to treatment. As per a study published in the Lancet in 2022, the gut microbiome of patients with metastatic melanoma can influence their response to immune checkpoint inhibitors. The study identified specific microbial species that were associated with better response rates and longer progression-free survival, suggesting that microbiome-targeted interventions could enhance the efficacy of cancer immunotherapies.

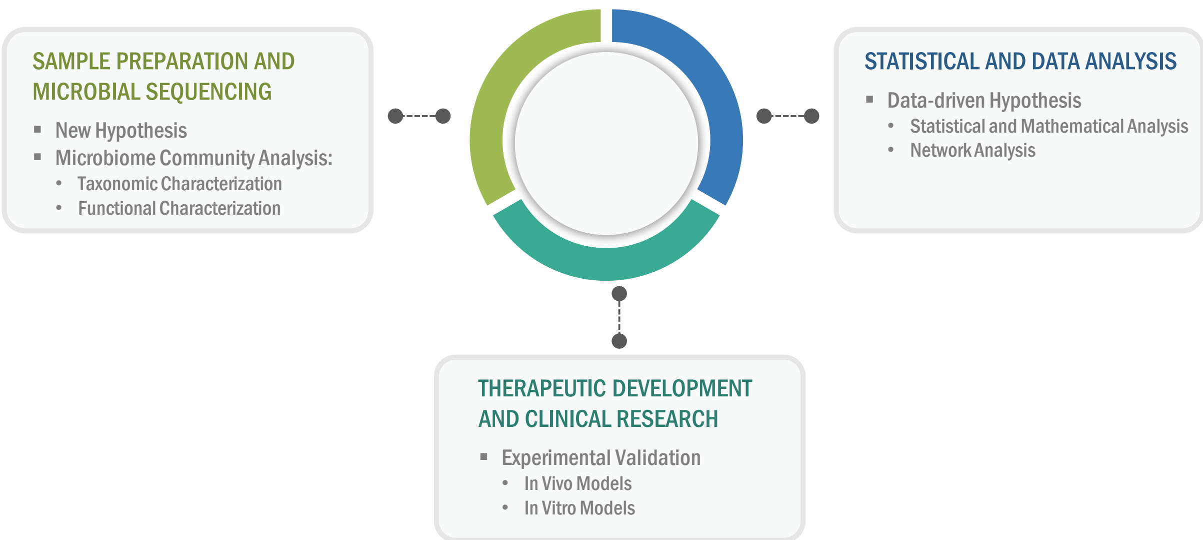
The demand for metagenomic sequencing has been driven by its applications in precision medicine, gut-brain axis research, and recent developments in sequencing technologies and data analysis methods. Metagenomic sequencing is proving to be a valuable tool for studying complex microbial communities and understanding their roles in various fields of research. The emergence of metagenomics has resolved the challenge of capturing the entire microbial community in different sample types by enabling the analysis of whole genomes without the need to culture the samples first. This is expected to boost the growth of the microbiome sequencing services market.

5.2.1.2 Growing focus on human microbiome therapeutics development

The increasing focus on human microbiome therapeutics development is driving therapeutics manufacturers to opt for sequencing, sample preparation, and data analysis services. This is further propelled by the need for a deeper understanding of the microbiome, including its composition, function, and interaction with the host, to develop microbiome-based therapeutics. Microbiome sequencing plays a critical role in providing this understanding by allowing for the identification and characterization of the microorganisms that make up the microbiome. Medium-sized and small-scale therapeutic manufacturers often prefer to outsource the initial stages of microbiome therapeutics development, such as screening, sample preparation, sequencing, and data analysis. This has offered significant momentum to large-scale CROs and service providers to develop their sequencing portfolio.

Human gastrointestinal microbiome research has the potential to offer pivotal clinical insights and facilitate the therapeutic development workflow. A research study conducted in June 2021, funded by the Australian National Health and Medical Research Council, highlighted a suitable avenue to translate microbiome research workflow, including the compositional and functional characterization of the microbiome, data-driven hypotheses generation, and the experimental validation of the hypotheses. By integrating different sequencing technologies (such as shotgun, metagenomic, and multi-omics) and data processing solutions, the ability to identify causative agents and design the appropriate microbiome therapeutics in disease treatment is expected to accelerate. The diagram below highlights the role of microbiome sequencing in such workflows.

FIGURE 19 ROLE OF MICROBIOME SEQUENCING IN THERAPEUTICS DEVELOPMENT



Source: Secondary Literature and MarketsandMarkets Analysis

Over the last few years, microbiome-based therapeutics development has witnessed significant advancements ranging from prebiotics and probiotics development to live biotherapeutics and microbiome mimetics. Advancements in bacterial culture methods from gastrointestinal (GI) samples, along with the expanding applications of metagenomic sequencing, have overcome several technical limitations in microbiome-based therapeutics development. However, a persisting challenge in this area is the development of appropriate methods to identify, refine, and test candidate therapies for microbiome medicines. Market players are progressing toward optimizing methods to identify candidate microbes, develop appropriate preclinical validation models, and progress to the personalized targeting of microbiome-based medicines through sequencing capabilities. This is expected to propel the market growth.

5.2.1.3 High cost of advanced infrastructure for sequencing

The field of microbiome sequencing requires sophisticated and expensive equipment, such as next-generation sequencing (NGS) platforms, bioinformatics tools, and data storage and analysis capabilities. Setting up and maintaining such infrastructure can be cost-prohibitive for many research and academic institutes, pharmaceutical and biotechnology companies, and other end users. Outsourcing microbiome sequencing to specialized service providers allows professionals to access advanced infrastructure and expertise without the need for substantial upfront investments. These service providers typically have state-of-the-art facilities and experienced personnel who are skilled in performing microbiome sequencing using the latest technologies and methodologies. Through outsourcing, professionals can leverage the capabilities of these specialized facilities without the burden of capital expenditures, operational costs, and maintenance expenses associated with in-house infrastructure.

The cost of NGS platforms is the highest in North America, followed by Europe. The cost in all regions is expected to increase by 2027 due to the high cost of raw materials, manufacturing, and labor. The cost of NGS platforms is low in the Middle East, Africa, and South America compared to Europe, North America, and the Asia Pacific due to the low cost of raw materials and the availability of a low-cost workforce (which reduces the overall manufacturing cost).

TABLE 5 AVERAGE SELLING PRICE OF NGS SYSTEMS, BY KEY PLAYER (2021)

COMPANY	INSTRUMENT	AVERAGE COST RANGE (USD)
Illumina	NovoSeq System	~850,000–900,000
Illumina	NextSeq System	~200,000–250,000
Illumina	MiSeq System	~80,000–85,0000
Illumina	iSeq System	~18,000–20,000
Illumina	Mini Seq System	~40,000–45,000
PacBio	Sequel System	~350,000–370,000
PacBio	Sequel IIe System	~490,000–500,000

Source: Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

5.2.2 RESTRAINTS

5.2.2.1 Dearth of skilled personnel

The shortage of skilled personnel with expertise in sequencing and data analysis is a significant challenge in the field of microbiome sequencing. Microbiome research involves complex methodologies, including sample collection, DNA extraction, library preparation, sequencing, and data analysis. The proper execution of these steps requires specialized knowledge and technical expertise. However, the field of microbiome sequencing is relatively new and evolving rapidly, with a shortage of skilled personnel with the necessary expertise. Training and developing skilled personnel in microbiome sequencing can be time-consuming and resource-intensive, and there may be a lag in keeping up with the latest advancements in sequencing technologies and data analysis methods.

5.2.3 OPPORTUNITIES

5.2.3.1 Expanding applications of microbiome sequencing in therapeutics development

Microbiome sequencing is expanding its applications beyond research and is increasingly being utilized in therapeutics development, including the study of the skin microbiome and oral microbiome sequencing. These applications have the potential to advance the understanding of the role of microbiota in health and disease and inform the development of targeted interventions for improving skin health and oral health.

Skin microbiome sequencing has the potential to unravel the complex interactions between the skin microbiome and the host immune system to identify key microbial species or functional genes associated with specific skin conditions and elucidate mechanisms underlying the therapeutic effects of probiotics or other interventions. It can help inform the development of novel therapeutic strategies, including probiotics, prebiotics, and targeted antimicrobial treatments aimed at modulating the skin microbiome to improve skin health. Additionally, the speedy expansion of gut microbiome research activities has provided a foundation to explore the role of microbiota in other physiological systems, including the skin. Skin microbiome sequencing is targeted toward understanding the temporospatial distribution of cutaneous microbial communities. Research on cutaneous microbial communities is anticipated to provide crucial insights into microbiota in skin aging and pathology. Microbiome sequencing technologies are set to gain significant demand to overcome the limited information on correlative analysis, host-microbe interactions in the skin, and low taxonomic resolution. This is expected to open new avenues of business expansion for service providers operating in this market.

Oral microbiome sequencing can provide insights into the microbial etiology of oral health conditions, identify key pathogenic species or functional genes associated with oral diseases, and reveal potential therapeutic targets for intervention. This knowledge can be used to develop novel strategies for oral health management, including probiotics, antimicrobial treatments, and personalized oral care regimens tailored to an individual's oral microbiome.

5.2.4 CHALLENGES

5.2.4.1 Issues related to sample preparation/isolation of microbes for sequencing

Despite the success of using metagenomic sequence data to guide the isolation and cultivation of microbes, there are still challenges and limitations that need to be addressed for more widespread success. Anaerobic microbes may also pose technical challenges for some targeted isolation methods due to limitations with cell sorters. Determining suitable growth media and physicochemical conditions for target organisms based solely on genome sequences can be challenging, as natural environment compositions are often unknown. Contamination can also be a significant challenge in microbiome research, as it can introduce non-target microbial DNA from external sources during sample collection, processing, and sequencing. Contamination can arise from reagents, equipment, and environmental sources, leading to false-positive or false-negative results. Proper controls, including negative controls and rigorous contamination prevention measures, should be implemented to minimize contamination and ensure accurate and reliable sequencing results.

Recovering genomic data from target microbes for cultivation strategies can also be impacted by issues such as incomplete cell extraction, cell lysis, or DNA degradation, which may affect the representation of certain species in metagenome data. DNA extraction is a critical step in sample preparation for microbiome sequencing, and different DNA extraction methods can yield different results. Some DNA extraction methods may be biased towards certain microbial groups or may not effectively lyse certain tough-to-break cells, resulting in the incomplete representation of the microbial community in the sequencing data.

5.3 TECHNOLOGY ANALYSIS

Sequencing by Synthesis (SBS)

Sequencing by synthesis (SBS) is a widely used technology in microbiome sequencing services for analyzing microbial communities. SBS is a high-throughput DNA sequencing method that enables the rapid and cost-effective sequencing of millions of DNA fragments in parallel. It has revolutionized the field of microbiome research by allowing researchers to explore the diversity and composition of microbial populations in each sample with high resolution.

The basic principle of SBS involves the use of DNA polymerases, enzymes that can synthesize DNA strands, to sequentially add nucleotides to a growing DNA chain. As each nucleotide is added, it is labeled with a fluorescent dye that corresponds to the base being incorporated. The fluorescence signal is then detected and recorded, allowing the identification of the nucleotide that was added. This process is repeated multiple times, typically in cycles, to generate DNA sequences that can be assembled into a complete microbial community profile.

Sequencing by Ligation (SBL)

Sequencing by ligation (SBL) is a DNA sequencing technology that is used in microbiome sequencing services for analyzing microbial communities. SBL is a high-throughput DNA sequencing method that relies on the ligation of short DNA oligonucleotide adapters to target DNA fragments, followed by their sequential hybridization and ligation to complementary oligonucleotides, enabling the determination of the DNA sequence.

The basic principle of SBL involves the use of DNA adapters that are ligated to the DNA fragments to be sequenced. These adapters contain known sequences at their ends that serve as priming sites for the sequencing reaction. The DNA fragments, together with the adapters, are then subjected to multiple rounds of hybridization and ligation with a set of fluorescently labeled oligonucleotides that are complementary to the known adapter sequences. The ligation events are detected and recorded, allowing the determination of the DNA sequence.